

HDOC Day-21: Menu-Driven C Program

Batches 15 & 16 | C Programming Activity

Aim

To develop a menu-driven C program that accepts two positive integers m and n ($m \leq n$) and uses a switch statement to display multiplication, addition, subtraction, and division tables for every integer from m to n . The menu must repeat until the user chooses Exit.

Problem Requirements

- Accept two positive integers m and n and ensure that $m \leq n$.
- Option 1: Print the first 10 multiples of every integer from m to n .
- Option 2: Print the addition table of every integer from m to n using addends 1 to 10.
- Option 3: Print the subtraction table of every integer from m to n using subtrahends 1 to 10.
- Option 4: Divide every integer from m to n by 1 to 10 and display each result rounded to two decimal places.
- Option 5: Exit the program.
- Display an error message for an invalid menu choice and continue showing the menu until Exit is selected.

Algorithm

1. Start.
2. Input positive integers m and n .
3. If $m \leq 0$, $n \leq 0$, or $m > n$, display an error message and ask for m and n again.
4. Display the menu with choices 1 to 5 and read the user choice.
5. Use a switch statement on the menu choice.
6. For choices 1 to 4, use an outer loop from m to n and an inner loop from 1 to 10 to print the selected table.
7. For division, convert the numerator to floating point and print the result using `%.2f`.
8. For an invalid choice, display an appropriate error message.
9. Repeat the menu until choice 5 is selected.
10. Stop.

C Program

```
#include <stdio.h>

int main()
{
    int m, n, choice, i, j;

    do {
        printf("Enter two positive integers m and n (m <= n): ");
        scanf("%d %d", &m, &n);

        if (m <= 0 || n <= 0 || m > n)
            printf("Invalid input. Please enter positive integers with m <= n.\n");
    } while (m <= 0 || n <= 0 || m > n);

    do {
        printf("\n----- MENU ----- \n");
        printf("1. Multiplication tables\n");
        printf("2. Addition tables\n");
        printf("3. Subtraction tables\n");
        printf("4. Division tables\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                for (i = m; i <= n; i++) {
                    printf("\nMultiplication table of %d:\n", i);
                    for (j = 1; j <= 10; j++)
                        printf("%d x %d = %d\n", i, j, i * j);
                }
                break;

            case 2:
                for (i = m; i <= n; i++) {
                    printf("\nAddition table of %d:\n", i);
                    for (j = 1; j <= 10; j++)
                        printf("%d + %d = %d\n", i, j, i + j);
                }
                break;

            case 3:
                for (i = m; i <= n; i++) {
                    printf("\nSubtraction table of %d:\n", i);
                    for (j = 1; j <= 10; j++)
                        printf("%d - %d = %d\n", i, j, i - j);
                }
                break;

            case 4:
                for (i = m; i <= n; i++) {
                    printf("\nDivision table of %d:\n", i);
                    for (j = 1; j <= 10; j++)
                        printf("%d / %d = %.2f\n", i, j, (float)i / j);
                }
                break;

            case 5:
                printf("Exiting program...\n");
        }
    }
}
```

```

        break;

    default:
        printf("Invalid menu choice. Please choose 1 to 5.\n");
    }
} while (choice != 5);

return 0;
}

```

Test-Case Table

Test	m, n	Choice	Expected Output	Actual / Result
1	2, 3	1	Multiplication tables of 2 and 3, each from x1 to x10	Passed - output matches expected multiplication tables.
2	2, 3	2	Addition tables of 2 and 3, using +1 to +10	Passed - output matches expected addition tables.
3	2, 3	3	Subtraction tables of 2 and 3, using -1 to -10	Passed - output matches expected subtraction tables.
4	2, 3	4	Division tables of 2 and 3, using /1 to /10, two decimals	Passed - division results are displayed to two decimal places.
5	2, 3	9	Invalid menu choice message; menu displayed again	Passed - invalid choice message displayed and menu repeated.
6	2, 3	5	Program exits successfully	Passed - Exit option terminates the program correctly.

Execution Evidence

The following terminal screenshots show successful compilation and execution of all menu choices using $m = 2$ and $n = 3$, including invalid-choice handling and program exit.

Screenshot 1 - Successful Compilation

```
(kali㉿kali)-[~/Desktop/apple]
$ gcc hdoc21.c -o hdoc21

(kali㉿kali)-[~/Desktop/apple]
$
```

Screenshot 2 - Multiplication Tables (Choice 1)

```
(kali㉿kali)-[~/Desktop/apple]
$ ./hdoc21
Enter two positive integers m and n ( $m \leq n$ ): 2
3

—— MENU ——
1. Multiplication tables
2. Addition tables
3. Subtraction tables
4. Division tables
5. Exit
Enter your choice: 1

Multiplication table of 2:
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

Multiplication table of 3:
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30
```

Screenshot 3 - Addition Tables (Choice 2)

```
——— MENU ———  
1. Multiplication tables  
2. Addition tables  
3. Subtraction tables  
4. Division tables  
5. Exit  
Enter your choice: 2
```

```
Addition table of 2:
```

```
2 + 1 = 3  
2 + 2 = 4  
2 + 3 = 5  
2 + 4 = 6  
2 + 5 = 7  
2 + 6 = 8  
2 + 7 = 9  
2 + 8 = 10  
2 + 9 = 11  
2 + 10 = 12
```

```
Addition table of 3:
```

```
3 + 1 = 4  
3 + 2 = 5  
3 + 3 = 6  
3 + 4 = 7  
3 + 5 = 8  
3 + 6 = 9  
3 + 7 = 10  
3 + 8 = 11  
3 + 9 = 12  
3 + 10 = 13
```

Screenshot 4 - Subtraction Tables (Choice 3)

```
——— MENU ———  
1. Multiplication tables  
2. Addition tables  
3. Subtraction tables  
4. Division tables  
5. Exit  
Enter your choice: 3
```

Subtraction table of 2:

```
2 - 1 = 1  
2 - 2 = 0  
2 - 3 = -1  
2 - 4 = -2  
2 - 5 = -3  
2 - 6 = -4  
2 - 7 = -5  
2 - 8 = -6  
2 - 9 = -7  
2 - 10 = -8
```

Subtraction table of 3:

```
3 - 1 = 2  
3 - 2 = 1  
3 - 3 = 0  
3 - 4 = -1  
3 - 5 = -2  
3 - 6 = -3  
3 - 7 = -4  
3 - 8 = -5  
3 - 9 = -6  
3 - 10 = -7
```

Screenshot 5 - Division Tables (Choice 4)

—— MENU ——

1. Multiplication tables
2. Addition tables
3. Subtraction tables
4. Division tables
5. Exit

Enter your choice: 4

Division table of 2:

2 / 1 = 2.00
2 / 2 = 1.00
2 / 3 = 0.67
2 / 4 = 0.50
2 / 5 = 0.40
2 / 6 = 0.33
2 / 7 = 0.29
2 / 8 = 0.25
2 / 9 = 0.22
2 / 10 = 0.20

Division table of 3:

3 / 1 = 3.00
3 / 2 = 1.50
3 / 3 = 1.00
3 / 4 = 0.75
3 / 5 = 0.60
3 / 6 = 0.50
3 / 7 = 0.43
3 / 8 = 0.38
3 / 9 = 0.33
3 / 10 = 0.30

Screenshot 6 - Invalid Menu Choice

```
—— MENU ——  
1. Multiplication tables  
2. Addition tables  
3. Subtraction tables  
4. Division tables  
5. Exit  
Enter your choice: 9  
Invalid menu choice. Please choose 1 to 5.  
—— MENU ——
```

Screenshot 7 - Exit (Choice 5)

```
—— MENU ——  
1. Multiplication tables  
2. Addition tables  
3. Subtraction tables  
4. Division tables  
5. Exit  
Enter your choice: 5  
Exiting program ...
```

Result

The menu-driven C program was compiled and executed successfully. Multiplication, addition, subtraction, and division tables for integers from m to n were produced correctly. Division results were rounded to two decimal places, an invalid menu choice displayed the appropriate error message, the menu continued to repeat, and the program terminated correctly when Exit was selected.